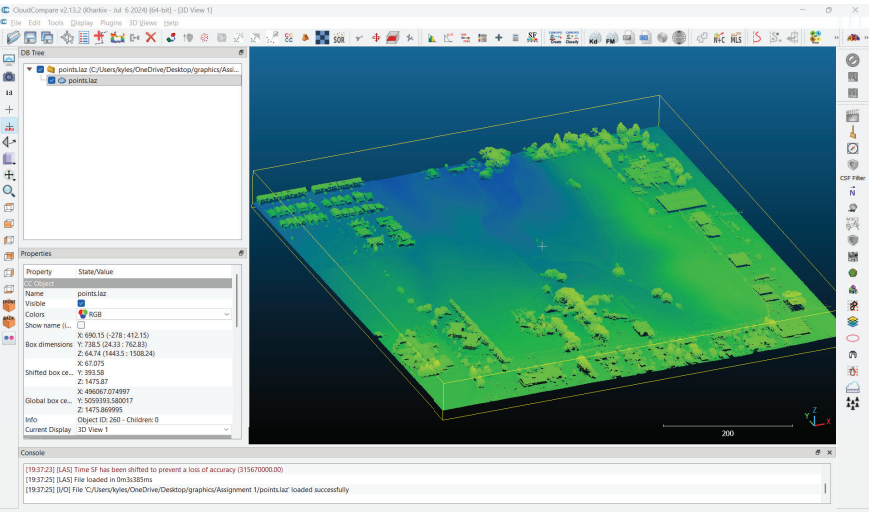
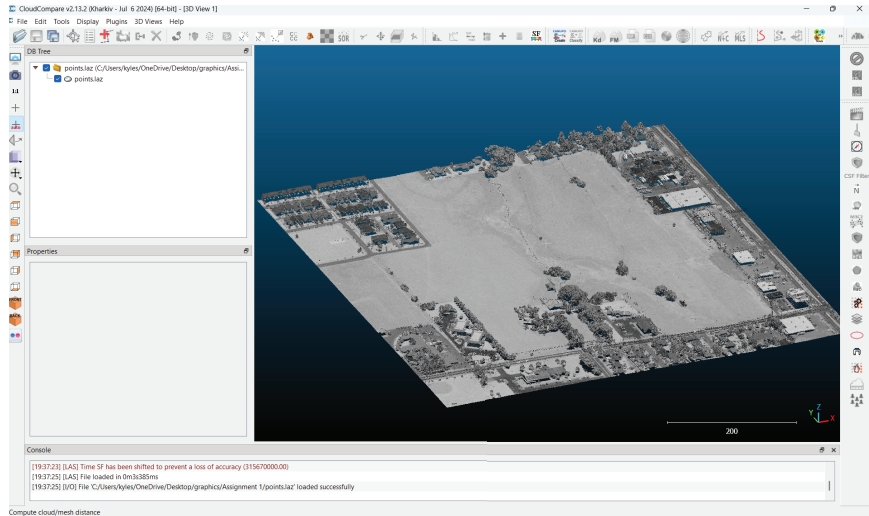


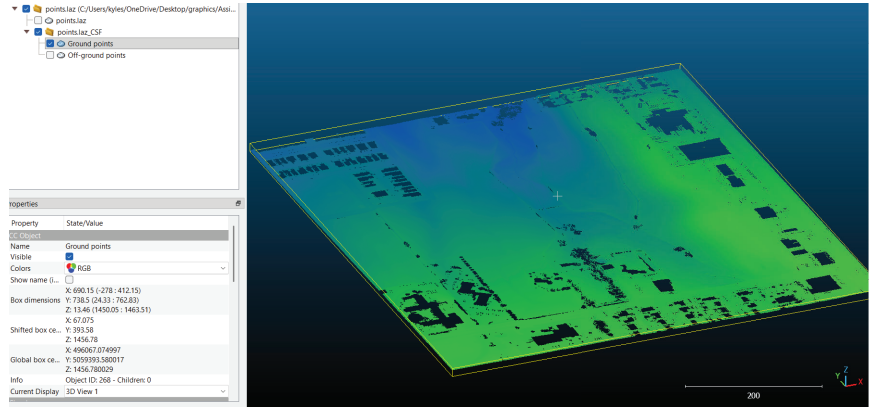
Topography

Cloud Compare

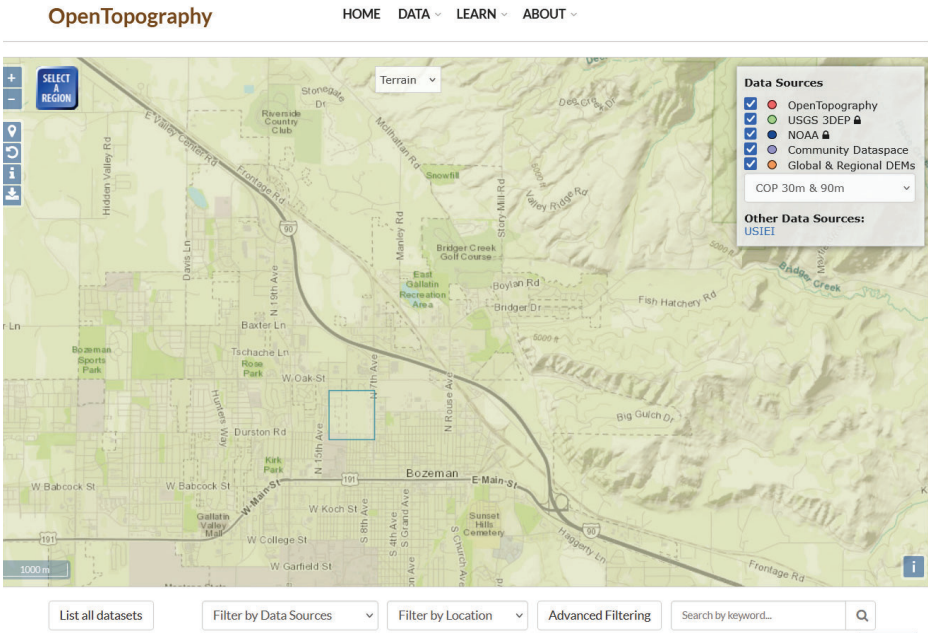
The CloudCompare program allows us to view the Lidar data into a 3D view. This lets us understand the data. Using a tool to recolor the data by a heatmap showing the depth of topography it shows a clearer visual of the form of the site



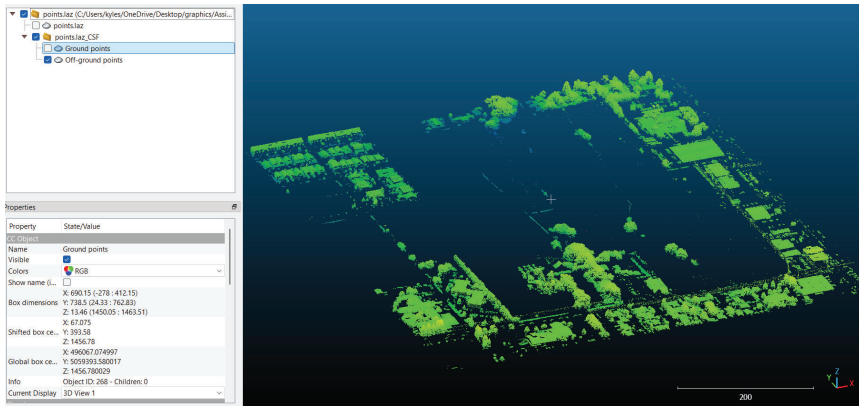
Taking this data we can further refine the information that is more critical for design process. This works through separating the ground point data from the above ground data. This ground point data is exported as a mesh to use in Rhino.



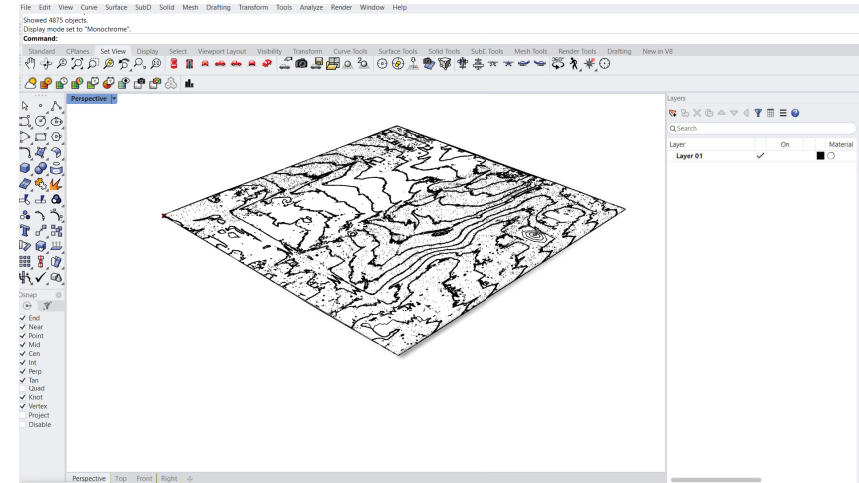
Using OpenTopography to download the point cloud data before importing the data into CloudCompare



The above ground data gives us information on the surroundings, such as buildings and foliage that may also be useful while creating topography in design projects



By creating a new file and opening the mesh from the ground points our topography shows in Rhino. From here we can create contour lines by the contour command and setting the height to one foot increments



Topography

Grasshopper

Another method to building topography is through the use of Grasshopper with the Heron package. This works by getting an earth anchor point using panels of latitude and longitude of where our site would be located. From there we can make bounding boxes by rectangles to set curves containers to reduce the size of the processing the program needs to do. We can get contours into a bounding box curve container with a number slider so that we can make the contour lines as close or far apart as needed combined with a color swatch so that we may be able to see the contour lines more clearly. Continuing on with the script we can make an actual topo solid and clean up the contours.

